Complex Engineering Problem for Embedded Systems (EE-354)

1. Project Title:

Smart Temperature and Humidity Monitoring System Using ATmega328P.

2. Abstract:

This project presents the design and implementation of a Smart Temperature and Humidity Monitoring System using the ATmega328P microcontroller. The system is developed to accurately measure ambient temperature and relative humidity using suitable digital sensors and to display the measured data in real time through a user-friendly display interface. The ATmega328P serves as the central processing unit, handling sensor data acquisition, signal processing, and peripheral control. The system is designed with an emphasis on reliability, low power consumption, and ease of integration, making it suitable for applications such as environmental monitoring, smart homes, laboratories, and industrial automation. Hardware components including sensors, display modules, and supporting circuitry are carefully selected and interfaced to ensure stable operation. The system architecture, working principle, and implementation methodology are discussed, along with performance evaluation under different environmental conditions. The results demonstrate accurate sensing, efficient data processing, and responsive display output. This project provides a practical and cost-effective solution for real-time temperature and humidity monitoring using an embedded system approach.

3. Design Objectives:

- Accurate real-time measurement of temperature and humidity.
- User-friendly display interface for visualizing measurements.
- Local and/or cloud-based data logging for monitoring and analysis.
- Modular, low-power, and reliable embedded system suitable for smart home or lab applications.
- Optional alert mechanism for threshold conditions.

4. Resource Identification

Human Resource (Project Team)

S.No.	Name	Roll No.	Section
1	Muhammad Taha*	EE-23319	D
2	Ali Nasir	EE-23328	D
3	Farzam Shaikh	EE-23326	B
4	Ahmed Abbas	EE-23325	B

Hardware Resource Identification

S. No.	Hardware Component/Equipment Name	Where/How to acquire
1	ATmega328P Microcontroller	-
2	SHT20/SHT21 Temperature & Humidity Sensor	-

3	16×2 Character OLED Module	-
4	Veroboard	-
5	Connecting wires	-
6	Buzzer	-
7	Battery pack (9V)	-

Software Resource Identification

S. No.	Software Name	Where/How to acquire
1	Code::Block	Official website: https://www.codeblocks.org/ (free download)
2	Arduino IDE	Official website: https://www.arduino.cc (free download)
3	Proteus	Official website: https://www.labcenter.com/ (free download)
4	Tinkercad	Official website: https://www.tinkercad.com/ (free download)

5. Project Objective Statement

The objective of this project is to propose and plan the development of a smart temperature and humidity monitoring system based on the ATmega328P microcontroller. The project aims to study suitable temperature and humidity sensors, identify appropriate hardware and software resources, and design a preliminary system architecture for real-time environmental monitoring.

The focus at this stage is on understanding sensor interfacing techniques, microcontroller-based data acquisition, and display integration through simulation and initial design analysis. The project also intends to evaluate the feasibility of adding features such as alert mechanisms and data logging in later stages. This preliminary phase establishes the technical foundation and design direction for the complete implementation of the system in subsequent phases.

6. Project Deliverables

The following deliverables are expected to be completed and submitted as part of the **preliminary phase** of the project:

1. **Project Proposal and Design Documentation**
   A detailed preliminary report including project objectives, literature review, system overview, block diagram, resource identification, and planned methodology.

2. **Preliminary System Architecture and Block Diagram**
   A conceptual block diagram showing the interaction between the ATmega328P, temperature and humidity sensor, display module, buzzer, and power supply.

3. **Initial Circuit Design and Simulation Plan**

Draft circuit schematics and a proposed simulation setup using tools such as Proteus or Tinkercad to validate sensor interfacing and display functionality.

4. **Software Design Strategy**
   An outline of the embedded software structure, including flowcharts or pseudocode describing sensor data acquisition, processing, display control, and alert logic.

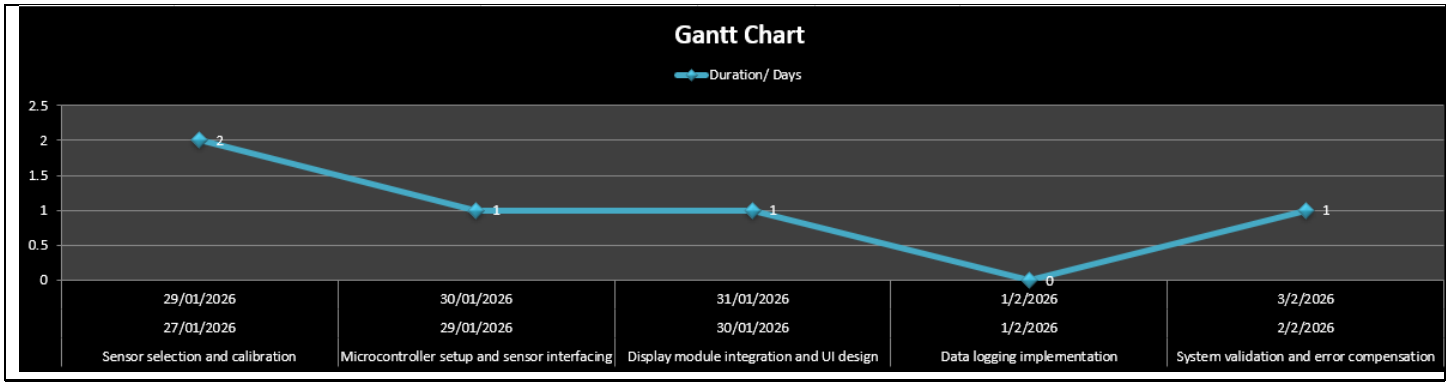
5. **Project Timeline and Gantt Chart**
   A planned timeline and Gantt chart defining project phases, task distribution, and expected milestones for successful project completion.

7. Resource Allocation and Time Scheduling

Hardware/Software Resource Allocation

Task	Start	End	Duration/ Days	Status
Sensor selection and calibration	27/01/2026	29/01/2026	2	Completed
Microcontroller setup and sensor interfacing	29/01/2026	30/01/2026	1	Completed
Display module integration and UI design	30/01/2026	31/01/2026	1	Completed
Data logging implementation	1/2/2026	1/2/2026	0	Completed
System validation and error compensation	2/2/2026	3/2/2026	1	Completed
Optional alert mechanism	3/2/2026	4/2/2026	1	Completed

Time Scheduling



9. Risk Assessment

Risk Assessment

1. **Sensor Malfunction or Inaccuracy**
   - Temperature and humidity sensors may give incorrect readings. Mitigation: Calibrate sensors and keep spare units.

2. **Microcontroller Failure**
   - ATmega328P may fail due to static or power issues. Mitigation: Use proper precautions and backup microcontroller.

3. **Communication/Interfacing Errors**
   - I2C connection between sensor and microcontroller may fail. Mitigation: Follow datasheet, test modules individually, and simulate connections.

4. **Power Supply Instability**
   - Unstable voltage can affect system operation. Mitigation: Use regulated supply and filtering capacitors.

5. **Software Bugs**

- Code errors can cause malfunction of data acquisition or display. Mitigation: Test code modules separately and simulate logic before hardware implementation.

10. References

[1] Sensirion AG, *SHT20 - Digital Humidity Sensor Datasheet*, Datasheet, Rev. A2, 2014. [Online]. Available: https://sensirion.com/media/documents/CCDE1377/635000A2/Sensirion_Datasheet_Humidity_Sensor_SHT20.pdf

[2] Sensirion AG, *SHT21 - Digital Humidity Sensor Datasheet*, Datasheet, Rev. 94, 2014. [Online]. Available: https://sensirion.com/media/documents/120BBE4C/63500094/Sensirion_Datasheet_Humidity_Sensor_SHT21.pdf

[3] Microchip Technology Inc., *ATmega328 - 8-bit AVR Microcontroller Datasheet*. [Online]. Available: https://en.wikipedia.org/wiki/ATmega328

[4] M. T. Ansari, *Humidity and Temperature Sensor Project Repository*, GitHub, 2026. [Online]. Available: https://github.com/Muhammad-Taha-Ansari/Humidity-And-Temperature-Sensor

Submission at the time of Evaluation

Skill(s) to be assessed	Extent of Achievement			
	50%	65%	80%	100%
Resource identification and allocation	Unable to identify and allocate	Partial identification and allocation	Partial identification or Partial allocation	Complete identification and allocation
Task identification and scheduling	Unable to identify and schedule	Partial identification and poor scheduling	Partial identification or Poor scheduling	Complete identification and proper scheduling
Bill of Materials (BOM) and budgeting	No BOM and budgeting	Incomplete BOM and budgeting	Complete BOM and budgeting with errors	Complete BOM and budgeting with no errors
Risk assessment and management	Unable to identify risks and unable to propose mitigation plan	Identifies risks correctly but unable to propose mitigation plan	Partial risk identification and partial mitigation plan proposed	Comprehensive risk assessment and mitigation plan presented

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